Learned Video Rate Control Under a Bit Budget: How Much Lookahead Does RL Need?

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Project idea

Every video encoder must decide how many bits each frame gets so that a segment of video (a group of pictures, GOP) fits a **hard bit budget** while looking as good as possible. The encoder controls a quantizer (QP) per frame: a lower QP spends more bits for a sharper frame, a higher QP spends fewer. This is a natural RL problem: the **state** is the budget left, recent QPs and the complexity of recent frames, plus the complexity of the next k frames (a lookahead window); the **action** is the next frame's QP; the **reward** is picture quality (PSNR/VMAF); and the **constraint** is that total bits must not exceed the budget, which is only revealed at the end of the episode. Spending too much early forces crushed quality later, and a frame's cost depends on how well the frames it references were coded, so the problem is neither myopic nor easy to model.

Research question. An offline search that sees a whole GOP in advance and tries various plans can beat the encoder's standard rate control, but it needs many trial encodes and cannot run in a live encoder. **How much of the hindsight search's benefit can a policy capture in a single encode, and how does that change with how many future frames it can see?** We measure the share of the search's gain captured as the lookahead grows from zero (strictly causal) to the full GOP. This gives live-encoder designers a concrete answer to what each extra frame of delay buys.

Approach. We work in real x264/x265 encoders with a fixed per-GOP budget, a controlled setting where exact comparisons are possible; long-video, buffer-based control is future work. (1) The hindsight search is the yardstick; baselines are constant-QP without adaptive quantization, the encoder's CRF and two-pass modes, and a feedback controller. (2) We compare RL algorithms trained in the real encoder under the hard budget: policy-gradient (PPO, A2C), value-based (DQN) and memory-based (RecurrentPPO) learners, and, for the best learner, a fixed penalty, a Lagrangian (dual-ascent) constraint and a safety shield for the budget. (3) The lookahead window is swept on the best learner, with the other algorithms checked at the endpoints. (4) Extensions, run in parallel and not required for the milestone: a privileged-teacher actor-critic whose critic sees the whole GOP while the deployed policy sees only its window, and a learned, uncertainty-aware simulator to cut real-encoder cost, motivated by our observation that policies tend to exploit simulator errors.

Milestone goals

Weeks	Goal	What we will do	How we measure it
1–2	Data, baselines, upper bound	Assemble a varied clip set (motion-heavy and static) with a held-out split. Set up baselines: constant-QP (no adaptive quantization), CRF, two-pass, feedback controller. Run the hindsight search for the upper bound. Time the real-encoder training loop.	Bits saved at equal quality (PSNR and VMAF); fraction of GOPs over budget; confidence intervals over GOPs.
2–3	Algorithm comparison (strictly causal)	Train PPO, A2C, DQN and RecurrentPPO in the real encoder with no lookahead. Pick the best learner; compare fixed penalty and Lagrangian budget handling on it.	Share of the search's gain captured per algorithm; overshoot rate versus quality lost; variation across random seeds.
3–4	Lookahead sweep	Retrain the best learner across lookahead windows from none to the full GOP; add the safety shield; check the other algorithms at the no-lookahead and full-GOP endpoints. Start the privileged-teacher variant in parallel.	Share of the search's gain captured versus lookahead window.
4–5	Evaluation and write-up	Score all policies on held-out clips in the real encoder. Report the lookahead curve and the overshoot tradeoff, whether positive or negative, with a diagnosis of where gains and losses come from.	Success criteria set from the measured upper bound and seed-to-seed variation (Weeks 1–2), before training; results with intervals, including negative results.